



МИНОБРНАУКИ РОССИИ

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технологий (МОСИТ)

**ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ № 3**

**по дисциплине**

**«Тестирование и верификация программного обеспечения»**

**Тема: «Подходы для разработки программного обеспечения»**

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## 1. ПОСТАНОВЛЕНИЕ ЗАДАЧИ

**Цель работы:** изучение и применение различных подходов к разработке программного обеспечения, основанного на тестировании, для повышения качества, надёжности и поддерживаемости кода.

### **Задачи:**

- 1) изучить теоретические основы методологий тестирования: TDD, ATDD, BDD и SDD;
- 2) исследовать преимущества и недостатки каждого подхода;
- 3) реализовать практические примеры для каждого метода;
- 4) проанализировать влияние интеграции тестирования на архитектуру и качество программного продукта;
- 5) подготовить итоговый отчёт с выводами и рекомендациями по интеграции подходов в современные процессы разработки.

## 2. ПРАКТИЧЕСКАЯ ЧАСТЬ

### 2.1 Реализация TDD

```
1  extends GUTest
2
3
4  var simple_video_container_scene: PackedScene = preload("path")
5  var viedo_player_scene: PackedScene = preload("path")
6
7  var simple_video_container: SimpleVideoContainer
8
9
10 func _ready():
11     >| viedo_player: VideoPlayer = viedo_player_scene.instantiate()
12     >|
13     >| get_tree().current_scene.add_child(viedo_player)
14
15
16 func test_play_wvd() -> void:
17     >| viedo_player_scene.set_video("path_to_wvd")
18     >|
19     >| assert_eq(viedo_player_scene._on_play_button_pressed, 0, "Video player mast have setted video stream")
20
21
22 func test_play_piano() -> void:
23     >| viedo_player_scene.set_video("path_to_piano")
24     >|
25     >| assert_eq(viedo_player_scene._on_play_button_pressed, 0, "Video player mast have setted video stream")
26
```

Рисунок 2.1.1 – Создание тестов часть 1

```
1  extends GUTest
2
3
4  var simple_video_container_scene: PackedScene = preload("path")
5  var viedo_player_scene: PackedScene = preload("path")
6
7  var simple_video_container: SimpleVideoContainer
8
9
10 func _ready():
11     >| viedo_player: VideoPlayer = viedo_player_scene.instantiate()
12     >|
13     >| get_tree().current_scene.add_child(viedo_player)
14
15
16 func test_search_look_outside() -> void:
17     >| assert_eq(viedo_player_scene.search("look_outside_name"), ["look_outside"],
18     >| >| >| >| >| >| >| >| >| "Video player mast have setted video stream")
19
20
21 func test_play_piano() -> void:
22     >| assert_eq(viedo_player_scene.search("look_outside_name"), ["piano_1", "piano_2"],
23     >| >| >| >| >| >| >| >| >| "Video player mast have setted video stream")
24
```

Рисунок 2.1.2 – Создание тестов часть 2

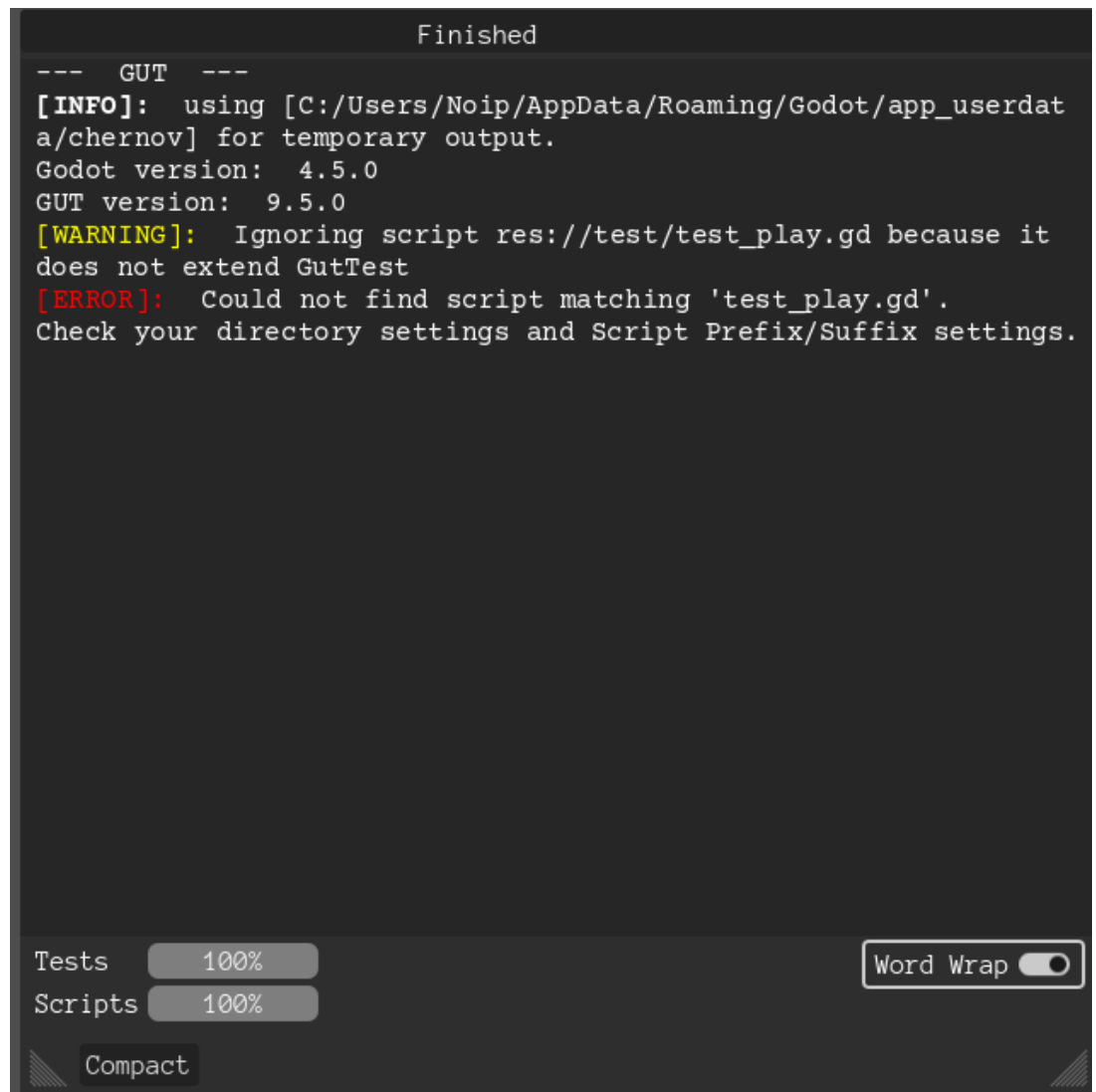


Рисунок 2.1.3 – Проведение тестов с инициализацией сцены

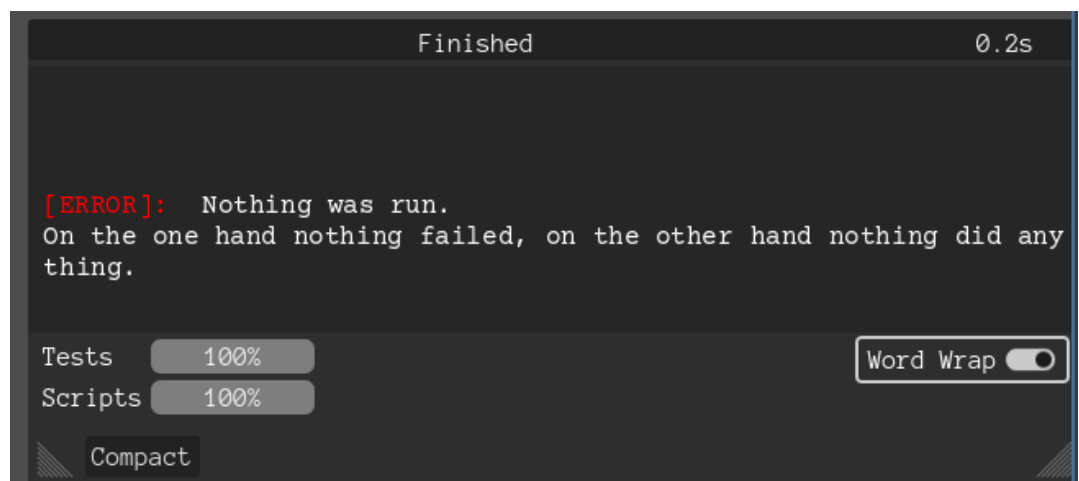


Рисунок 2.1.4 – Проведение тестов с без инициализации сцены

```

1  extends GutTest
2
3
4  var simple_video_container_scene: PackedScene = preload("res://scenes/simple_video_container.tscn")
5  var video_player_scene: PackedScene = preload("res://scenes/video_player.tscn")
6
7  var simple_video_container: SimpleVideoContainer
8  var video_player: VideoPlayer
9
10 func _ready():
11     video_player = video_player_scene.instantiate()
12
13     get_tree().current_scene.add_child(video_player)
14     super()
15
16
17 func test_play_wwd() -> void:
18     video_player.set_video("War-Without-Drumming-Full_.ogv")
19
20     assert_eq(video_player._on_play_button_pressed(), 0, "Video player mast have setted video stream")
21
22
23 func test_play_piano() -> void:
24     video_player.set_video("Daten-Yofukashi-no-uta-0p-PIANO-TUTORIAL-FREE-SHEET.ogv")
25
26     assert_eq(video_player._on_play_button_pressed(), 0, "Video player mast have setted video stream")
27

```

Рисунок 2.1.5 – Завершённые тесты часть 1

```

1  extends GutTest
2
3
4
5  var simple_video_container_scene: PackedScene = preload("res://scenes/simple_video_container.tscn")
6  var video_player_scene: PackedScene = preload("res://scenes/video_player.tscn")
7
8  var simple_video_container: SimpleVideoContainer
9  var video_player: VideoPlayer
10
11 func _ready():
12     video_player = video_player_scene.instantiate()
13
14     get_tree().current_scene.add_child(video_player)
15     super()
16
17
18 func test_search_look_outside() -> void:
19     assert_eq(video_player.search("Gone"), ["Gone-Angels-on-Piano-Library-of-Ruina-_libraryofruina_mili-Lobotomycorporation-_limbusco
20     |> |> |> |> |> |> |> |> "Video player mast have setted video stream")
21
22
23 func test_play_piano() -> void:
24     print(video_player.search("p"))
25     assert_eq(video_player.search("p"), ["Daten-Yofukashi-no-uta-0p-PIANO-TUTORIAL-FREE-SHEET.ogv", "Gone-Angels-on-Piano-Library-of-Ruin
26     |> |> |> |> |> |> |> |> "kim-kitsuragi-is-not-really-impressed.ogv"],
27     |> |> |> |> |> |> |> |> "Video player mast have setted video stream")
28

```

Рисунок 2.1.6 – Завершённые тесты часть 2

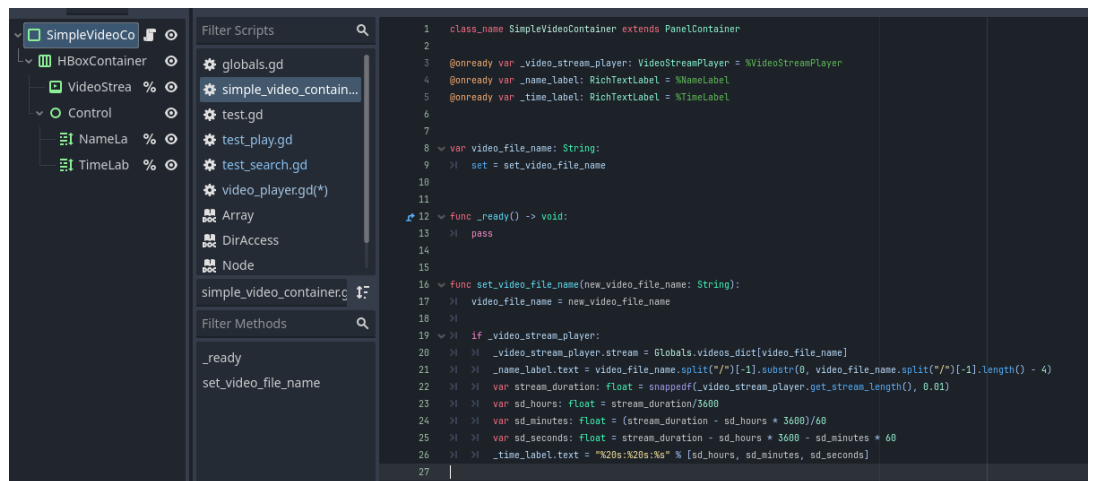


Рисунок 2.1.7 – Реализация функционала часть 1

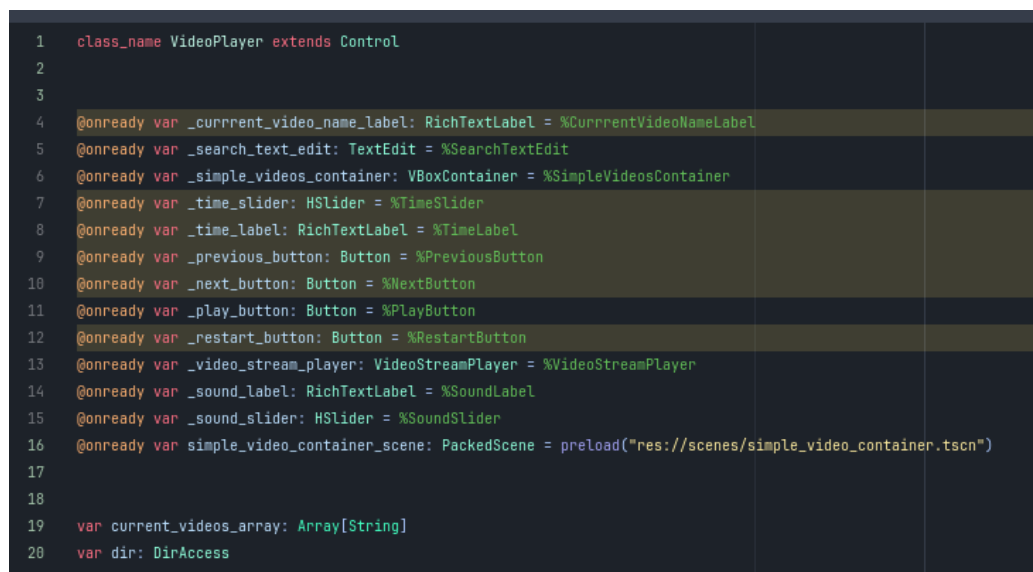


Рисунок 2.1.8 – Реализация функционала часть 2



Рисунок 2.1.9 – Реализация функционала часть 3

```

45 func prepare_global_video_dict() -> void:
46     >| var dir = DirAccess.open(Globals.videos_dict_folder_path)
47     >|
48     >| for file_name in dir.get_files():
49     >| >| if file_name.substr(file_name.length() - 4) == ".ogv":
50     >| >| >| Globals.videos_dict[file_name] = load(Globals.videos_dict_folder_path + file_name)
51
52
53 func set_video(video_name: String) -> void:
54     >| _video_stream_player.stream = Globals.videos_dict[video_name]
55     >|
56     >| _video_stream_player.play()
57
58
59 func search(mask: String = "") -> Array[String]:
60     >| var result: Array[String]
61     >| current_videos_array.clear()
62     >|
63     >| _clear_simple_videos_container()
64     >|
65     >| for video_name in Globals.videos_dict.keys():
66     >| >| if mask in video_name or mask == "":
67     >| >| >| result.append(video_name)
68     >| >| >| current_videos_array.append(video_name)
69     >| >| >|
70     >| >| >| var simple_video_container = _create_simple_video_container(video_name)
71     >| >| >| _simple_videos_container.add_child(simple_video_container)
72     >|
73     >| return result

```

Рисунок 2.1.11 – Реализация функционала часть 4

```

76 func _clear_simple_videos_container() -> void:
77     >| for child: Node in _simple_videos_container.get_children():
78     >| >| child.queue_free()
79
80
81 func _on_play_button_pressed() -> int:
82     >| if _video_stream_player.stream == null:
83     >| >| return -1
84     >|
85     >| _video_stream_player.paused = not _video_stream_player.paused
86     >|
87     >| if not _video_stream_player.paused:
88     >| >| _video_stream_player.play()
89     >|
90     >| _play_button.text = "Play" if _video_stream_player.paused else "Stop"
91     >|
92     >| return 0
93
94
95 func _on_reset_button_pressed() -> void:
96     >| pass
97
98
99 func _create_simple_video_container(video_name: String) -> SimpleVideoContainer:
100     >| var s_v_c: SimpleVideoContainer = simple_video_container_scene.instantiate()
101     >| s_v_c.set_video_file_name(video_name)
102     >| return s_v_c

```

Рисунок 2.1.12 – Реализация функционала часть 5



```

104
105 func _sound_value_hanged(sound_value: float) -> void:
106     >I var sound_value_db: float = linear_to_db(sound_value)
107     >I AudioServer.set_bus_volume_db(0, sound_value_db)
108     >I
109     >I _sound_label.text = "Громкость звука: %s" % (sound_value * 100)
110

```

Рисунок 2.1.13– Реализация функционала часть 6

```

Finished 3.4s

* test_play_piano
2/2 passed.

res://test/test_search.gd
* test_search_look_outside
* test_play_piano
2/2 passed.

=====
= Run Summary
=====

Totals
-----
Scripts          2
Tests            4
Passing Tests    4
Asserts          4
Time             3.421s

---- All tests passed! ----

Tests 100% res://test
Scripts 100% test_search.gd

Word Wrap
Compact

```

Рисунок 2.1.14 – Результаты тестов

## 2.2 Реализация ATDD

```

1  extends GutTest
2
3  var simple_video_container_scene: PackedScene = preload("res://scenes/simple_video_container.tscn")
4  var video_player_scene: PackedScene = preload("res://scenes/video_player.tscn")
5
6  var simple_video_container: SimpleVideoContainer
7  var video_player: VideoPlayer
8
9  func _ready():
10     video_player = video_player_scene.instantiate()
11
12     get_tree().current_scene.add_child(video_player)
13     super()
14
15
16  func test_selected_wwd() -> void:
17     video_player.search("War-Without-Drumming-Full.ogv")
18
19     assert_eq(video_player._simple_videos_container.get_child(0)._set_video_as_main(), 0, "video must be set and played")
20
21
22  func test_selected_kim() -> void:
23     video_player.search("kim-kitsuragi-is-not-really-impressed.ogv")
24
25     assert_eq(video_player._simple_videos_container.get_child(0)._set_video_as_main(), 0, "video must be set and played")
26

```

Рисунок 2.2.1 – Написание тестов часть 1

```

Finished 1.7s

res://test/test_play_selected.gd
* test_selected_wwd
[Failed]: Unexpected Errors:
[1] <engine-2>Trying to assign value of type 'PanelContainer' to a variable of type 'simple_video_container.gd'.
[2] <engine-2>Invalid access to property or key 'set_video_as_main' on a base object of type 'Nil'.
[3] <engine-0>Index p_index = 0 is out of bounds ((int)data.children_cache.size() - data.internal_children_front_count_cache - data.internal_c
hildren_back_count_cache = 0).
[4] <engine-2>Attempt to call function '_set_video_as_main' in base 'null instance' on a null instance.
at line -1
8 Orphans
* [SimpleVideoContainer:<PanelContainer#63518541647>] + 5
* test_selected_kim
[Failed]: Unexpected Errors:
[1] <engine-2>Trying to assign value of type 'PanelContainer' to a variable of type 'simple_video_container.gd'.
[2] <engine-2>Invalid access to property or key 'set_video_as_main' on a base object of type 'Nil'.
[3] <engine-0>Index p_index = 0 is out of bounds ((int)data.children_cache.size() - data.internal_children_front_count_cache - data.internal_c
hildren_back_count_cache = 0).
[4] <engine-2>Attempt to call function '_set_video_as_main' in base 'null instance' on a null instance.
at line -1
8 Orphans
* [SimpleVideoContainer:<PanelContainer#64290293625>] + 5
16 orphans
0/2 passed.

=====
= 24 Orphans
=====
res://test/test_play_selected.gd
- Outside Tests
* [SimpleVideoContainer:<PanelContainer#63048779573>] + 5
test_play_selected.gd
- test_selected_wwd
* [SimpleVideoContainer:<PanelContainer#63518541647>] + 5
- test_selected_kim
* [SimpleVideoContainer:<PanelContainer#64290293625>] + 5

=====
Tests 100% res://test
Scripts 100% test_play_selected.gd
Word Wrap
Compact

```

Рисунок 2.2.2 – Проведение тестов часть 2

```
Finished 1.7s

test_play_selected.gd
- test_selected_wvd
  * [SimpleVideoContainer:<PanelContainer#63518541647>] + 5
- test_selected_kim
  * [SimpleVideoContainer:<PanelContainer#64290293625>] + 5

=====
= Run Summary
=====

res://test/test_play_selected.gd
- test_selected_wvd
  [Failed]: Unexpected Errors:
  [1] <engine-2>Trying to assign value of type 'PanelContainer' to a variable of type 'simple_video_container.gd'.
  [2] <engine-2>Invalid access to property or key 'set_video_as_main' on a base object of type 'Nil'.
  [3] <engine-0>index p_index = 0 is out of bounds ((int)data.children_cache.size() - data.internal_children_front_count_cache - data.internal_c
hildren_back_count_cache = 0).
  [4] <engine-2>Attempt to call function '_set_video_as_main' in base 'null instance' on a null instance.
    at line -1
- test_selected_kim
  [Failed]: Unexpected Errors:
  [1] <engine-2>Trying to assign value of type 'PanelContainer' to a variable of type 'simple_video_container.gd'.
  [2] <engine-2>Invalid access to property or key 'set_video_as_main' on a base object of type 'Nil'.
  [3] <engine-0>index p_index = 0 is out of bounds ((int)data.children_cache.size() - data.internal_children_front_count_cache - data.internal_c
hildren_back_count_cache = 0).
  [4] <engine-2>Attempt to call function '_set_video_as_main' in base 'null instance' on a null instance.
    at line -1

Totals
-----
Scripts          1
Tests            2
Passing Tests     none
Failing Tests     2
Asserts           0/2
Orphans          24
Time             4.12s

Ran Scripts matching "test_play_selected.gd"
---- 2 failing tests ----

Tests 100% res://test
Scripts 100% test_play_selected.gd
Word Wrap
Compact
```

Рисунок 2.2.3 – Проведение тестов часть 3

```
Finished 0.3s

res://test/test_play_selected.gd
* test_selected_wvd
* test_selected_kim
2/2 passed.

=====
= Run Summary
=====

Totals
-----
Scripts          1
Tests            2
Passing Tests     2
Asserts           2
Time             0.31s

Ran Scripts matching "test_play_selected.gd"
---- All tests passed! ----

Tests 100% res://test
Scripts 100% test_play_selected.gd
Word Wrap
Compact
```

Рисунок 2.2.4 – Проведение тестов часть 4

```

39
40  signal set_video_as_main(video_name)]
41
42  func _set_video_as_main() -> int:
43  >| if video_file_name == "":
44  >| >| return -1
45  >|
46  >| set_video_as_main.emit(video_file_name)
47  >| return 0
48

```

Рисунок 2.2.5 – Реализация функционала часть 1

```

59  func search(mask: String = "") -> Array[String]:
60  >| var result: Array[String]
61  >| current_videos_array.clear()
62  >|
63  >| _clear_simple_videos_container()
64  >|
65  >| for video_name in Globals.videos_dict.keys():
66  >| >| if mask in video_name or mask == "":
67  >| >| >| result.append(video_name)
68  >| >| >| current_videos_array.append(video_name)
69  >| >| >|
70  >| >| >| var simple_video_container = _create_simple_video_container(video_name)
71  >| >| >| simple_video_container.set_video_as_main.connect(set_video)
72  >| >| >| _simple_videos_container.add_child(simple_video_container)
73  >|
74  >| return result

```

Рисунок 2.2.6 – Реализация функционала часть 2

## 2.3 Реализация BDD

|   |                                                                                                |  |
|---|------------------------------------------------------------------------------------------------|--|
| 1 | Scenario {check_played_wwd}                                                                    |  |
| 2 | Given the vidoes array of {Daten-Yofukashi-no-uta-0p-PIANO-TUTORIAL-FREE-SHEET.ogv}{Gone-Ar... |  |
| 3 | When user chooses video {War-Without-Drumming-Full_.ogv}                                       |  |
| 4 | Then chosen video starts play returning it's state {0}                                         |  |
| 5 |                                                                                                |  |

Рисунок 2.3.1 – Описание сценария часть 1

```

1  extends GutTest
2
3
4  enum BDD_TYPE{
5    >| SCENARIO,
6    >| GIVEN,
7    >| WHEN,
8    >| THEN,
9    >| NONE
10 }
11
12 var bdd_type_dict: Dictionary[BDD_TYPE, String] = {
13   >| BDD_TYPE.SCENARIO: "scenario",
14   >| BDD_TYPE.GIVEN: "given",
15   >| BDD_TYPE.WHEN: "when",
16   >| BDD_TYPE.THEN: "then",
17   >| BDD_TYPE.NONE: "none"
18 }
19
20
21 var simple_video_container_scene: PackedScene = preload("res://scenes/simple_video_container.tscn")
22 var video_player_scene: PackedScene = preload("res://scenes/video_player.tscn")
23
24 var simple_video_container: SimpleVideoContainer
25 var video_player: VideoPlayer
26 var scenario_values: Array = [[], [], [], []]
27

```

Рисунок 2.3.2 – Написание тестов часть 1

```

28 func _ready():
29   >| video_player = video_player_scene.instantiate()
30   >|
31   >| get_tree().current_scene.add_child(video_player)
32   >| super()
33
34
35 func _read_scenario() -> void:
36   >| var file: FileAccess = FileAccess.open("bdd/scenarios.txt", FileAccess.READ)
37   >| var current_line = "_"
38   >|
39   >| while true:
40     >| >| current_line = file.get_line()
41     >| >| if current_line != "":
42       >| >| >| var line_type: BDD_TYPE = _get_bdd_type(current_line.split(" ")[0])
43       >| >| >| var line_values: Array = _get_line_values(current_line)
44       >| >| >| scenario_values[line_type].append(line_values)
45       >| >| >|
46     >| >| >| if line_type > BDD_TYPE.SCENARIO:
47       >| >| >| >| print("%s%s" % [bdd_type_dict[line_type], scenario_values[0][0][0]], line_values)
48       >| >| >| >| call("%s%s" % [bdd_type_dict[line_type], scenario_values[0][0][0]], line_values)
49     >| >| >| else:
50       >| >| >| break
51

```

Рисунок 2.3.3 – Написание тестов часть 2

```

71 func test_selected_wwd() -> void:
72     >| _read_scenario()
73
74
75 func given_check_played_wwd(video_array: Array) -> void:
76     >| print(video_player.current_videos_array)
77     >| assert_eq(video_player.current_videos_array, video_array, "array of videos must equal")
78
79
80 func when_check_played_wwd(video_name: Array) -> void:
81     >| video_player.set_video(video_name[0])
82     >|
83     >| assert_eq(video_player._video_stream_player.stream.file, "res://videos/War-Without-Drumming-Full_.ogv", "f
84
85
86 func then_check_played_wwd(values_array: Array) -> void:
87     >| assert_eq(video_player._on_play_button_pressed(), int(values_array[0]), "video must be played correctly")
88
89

```

Рисунок 2.3.4 – Написание тестов часть 3

```

71
72 func _get_bdd_type(word: String) -> BDD_TYPE:
73     >| match word:
74         >| "Scenario":
75             >| return BDD_TYPE.SCENARIO
76         >| "Given":
77             >| return BDD_TYPE.GIVEN
78         >| "When":
79             >| return BDD_TYPE.WHEN
80         >| "Then":
81             >| return BDD_TYPE.THEN
82         >| _:
83             >| return BDD_TYPE.NONE
84
85
86 func _get_line_values(text_line: String) -> Array:
87     >| var values: Array
88     >|
89     >| var text_line_values: PackedStringArray = text_line.split("{")
90     >|
91     >| for value in text_line_values:
92         >| if value.length() != 0 and value[-1] == "}":
93             >| values.append(value.split("}")[0])
94     >|
95     >| return values
96

```

Рисунок 2.3.5 – Написание тестов часть 4

```
Finished 0.3s

res://test/test_play_selected.gd
* test_selected_wwd
  [Risky]: test_selected_wwd did not assert

=====
= Run Summary
=====
- test_selected_wwd
  [Risky]: Did not assert

Totals
-----
Scripts          1
Tests            1
Passing Tests    none
Risky/Pending    1
Asserts          none
Time             0.304s

Ran Scripts matching "test_play_selected.gd"
---- 1 pending/risky tests. ----

Tests 100% res://test
Scripts 100% test_play_selected.gd
Word Wrap
Compact
```

Рисунок 2.3.6 – Проверка работоспособности тестов

```
Finished 0.3s

res://test/test_play_selected.gd
* test_selected_wwd
1/1 passed.

=====
= Run Summary
=====

Totals
-----
Scripts          1
Tests            1
Passing Tests    1
Asserts          3
Time             0.317s

Ran Scripts matching "test_play_selected.gd"
---- All tests passed! ----

Tests 100% res://test
Scripts 100% test_play_selected.gd
Word Wrap
Compact
```

Рисунок 2.3.7 – Результат тестов

## 2.4 SDD

```
1  {
2      name : {
3          type: *extension type*,
4          duration: format(hh:mm:ss.mm),
5          playing: true/false
6      }
7  }
```

Рисунок 2.4.1 – Пример БД

```
1  extends GwtTest
2
3
4  var simple_video_container_scene: PackedScene = preload("res://scenes/simple_video_container.tscn")
5  var video_player_scene: PackedScene = preload("res://scenes/video_player.tscn")
6
7
8  var simple_video_container: SimpleVideoContainer
9  var video_player: VideoPlayer
10 var db_example: Dictionary
11
12 func _ready():
13     >| video_player = video_player_scene.instantiate()
14     >|
15     >| get_tree().current_scene.add_child(video_player)
16     >| super()
17
18
19 func test_db_wvd() -> void:
20     >| video_player.set_video("War-Without-Drumming-Full_.ogv")
21     >|
22     >| add_to_db(video_player._video_stream_player)
23     >| print(db_example)
24     >| assert_eq(db_example["War-Without-Drumming-Full"]["playing"], true, "Video must be playing")
```

Рисунок 2.4.2 – Написание тестов часть 1



```

26 ▾ func test_db_kim() -> void:
27   >| video_player.set_video("kim-kitsuragi-is-not-really-impressed.ogv")
28   >|
29   >| video_player._video_stream_player.stop()
30   >|
31   >| add_to_db(video_player._video_stream_player)
32   >| assert_eq(db_example["kim-kitsuragi-is-not-really-impressed"]["playing"], false, "Video must be paused")
33
34
35 ▾ func add_to_db(video_stream_player: VideoStreamPlayer) -> void:
36   >| var stream: VideoStreamTheora = video_stream_player.stream
37   >| var file_name: String = stream.file.split("/")[1].split(".")[0]
38   >| db_example[file_name] = {}
39   >| db_example[file_name]["type"] = stream.file.split(".")[1]
40   >| db_example[file_name]["duration"] = count_time(video_stream_player.get_stream_length())
41   >| db_example[file_name]["playing"] = video_stream_player.is_playing()
42
43
44 ▾ func count_time(time: float) -> String:
45   >| var stream_duration: float = snappedf(time, 0.01)
46   >| var sd_hours: int = stream_duration/3600
47   >| var sd_minutes: int = (stream_duration - sd_hours * 3600)/60
48   >| var sd_seconds: float = stream_duration - sd_hours * 3600 - sd_minutes * 60
49   >| return "%02d:%02d:%s" % [sd_hours, sd_minutes, sd_seconds]
50

```

Рисунок 2.4.3 – Написание тестов часть 2

```

Finished 0.3s

res://test/test_db_example.gd
* test_db_wvd
* test_db_kim
2/2 passed.

=====
= Run Summary
=====

Totals
-----
Scripts          1
Tests            2
Passing Tests    2
Asserts          2
Time             0.315s

Ran Scripts matching "test_db_example.gd"
---- All tests passed! ----

Tests 100% res://test
Scripts 100% test_db_example.gd

Word Wrap
Compact

```

Рисунок 2.4.4 – Результат тестов

### **3. АНАЛИЗ И ВЫВОДЫ**

Тестирования были проведены в полном объёме, а методы разработки - изучены.

В ходе работы были получены навыки по работе с TDD, ATDD, BDD, SDD. Полученные знания были закреплены путём выполнения практического задания.